Research to have a Proof of concept (Goal is to research and see “Can this acc be built/done”)

Project context

Client definition: WSP does engineering consulting. For example, a client may be The government and they want to build a new parliament building. They consult with WSP and WSP does the design and all the engineering stuff for the building. Here The client is the government

End User: WSP’s engineers (structural engineers) will be using this tool, not the most software technical people

The aim is to let owners and architects make decisions while the design can still be influenced.

(Before too late into project timeline where design changes can't be made anymore)

The process should be largely autonomous, with a human reviewing the results at key points.

The vision/ flow

1. The input: A basic building model

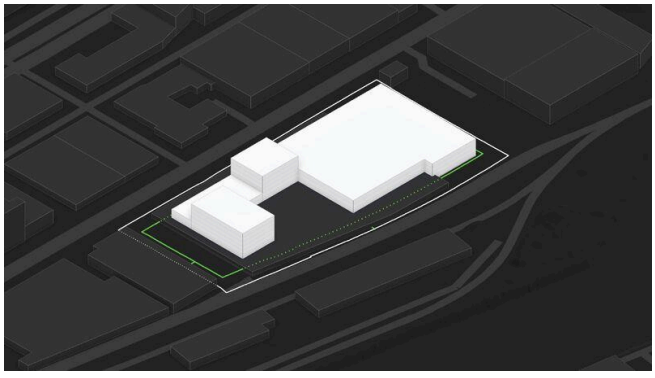
At this stage:

- It might only show the building's shape, size, height, and floors.
- It doesn't necessarily specify how the building will actually be supported or constructed.

2. Next: The AI agent generates different options

The client might say:

“I'm interested in concrete, steel or timber in this location; show me what it could look

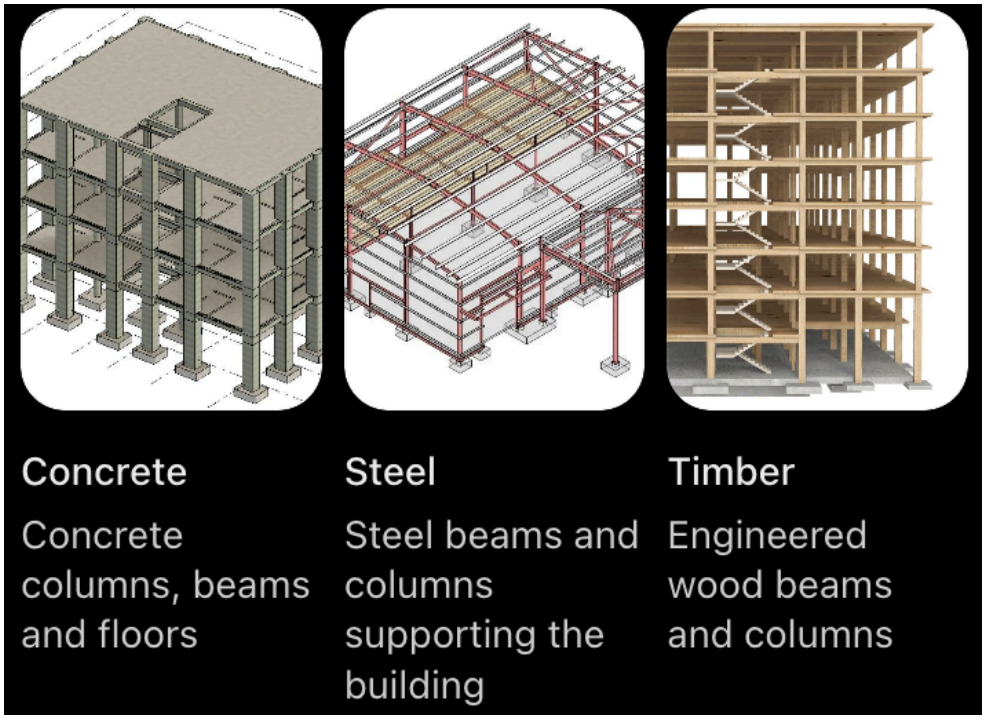


like.” The
agent iterates the designs, then builds and visualizes each option with its trade-offs in cost, embodied carbon, schedule and bay sizing.

or

"I want to build this office building in Toronto. Show me what it would look like using concrete, steel, or timber."

The agent would explore three possible structural systems:



The agent isn't just changing the appearance. It's exploring how each material could be used to support the same building*** -> would have to research how possible this is, or can it only be done at a way less optimal scale

3. The agent compares the options

For each design, the system produces preliminary estimates.

Factor	**What it means**
Cost	How expensive is it to construct?
Embodied carbon	How much greenhouse gas is associated with producing and using the construction materials?

Schedule	How long might construction take?
Bay sizing	How far apart can supporting columns be placed? This affects how much open floor space is available.

The estimates help architects and owners decide which design is worth developing further.

OUR MAIN SCOPE (What we have to make)

As just discussed (idek if im gonna leave this part in), for clarity this is our scope

Input (model)

↓

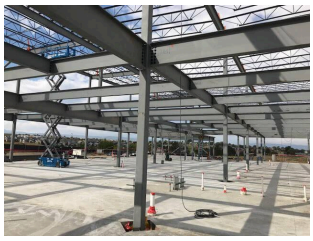
Agent processing (we have to build)

“The agent iterates structural framing options. Bay sizes are optimized for constructability and operations, per material (the optimal bay for timber differs from concrete).”

Structural framing options:

Imagine you're designing a building. You need columns and beams to hold it up.

There are multiple ways to arrange those columns and beams. The agent should explore different arrangements.

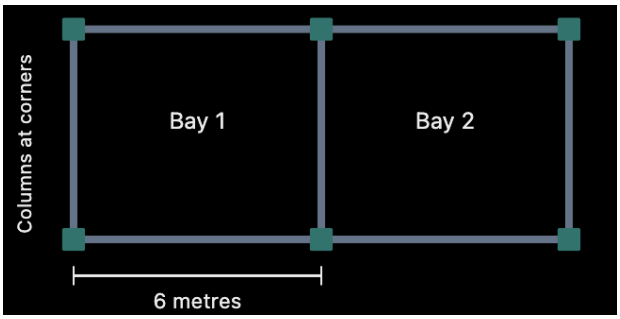



Bay Sizes:

A bay is the space between structural columns.

For example, the agent might compare:

- Option A: Columns spaced 4 metres apart.
- Option B: Columns spaced 6 metres apart.
- Option C: Columns spaced 8 metres apart.



Each option has different costs and engineering implications.

Material

Because different materials have different structural properties and construction methods.

For example:

- Timber: A 6-metre bay might work well with readily available timber beams.

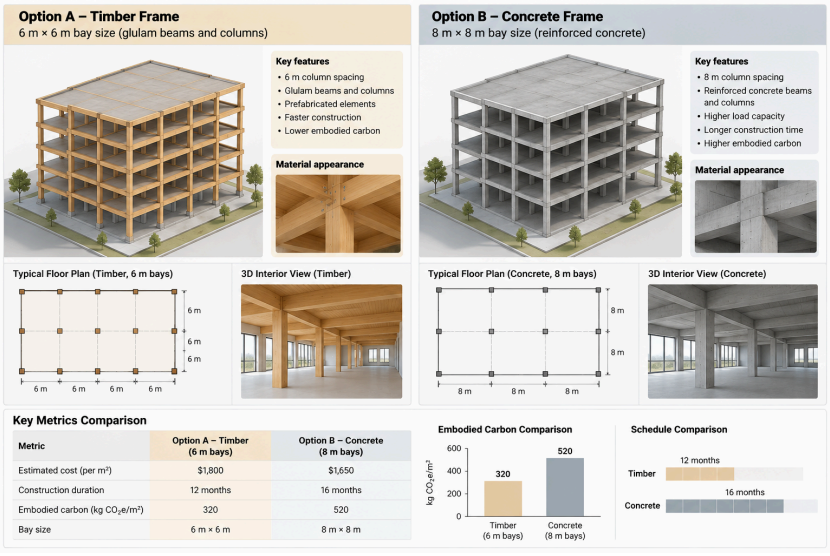
- Concrete: An 8-metre bay might be more economical for a particular concrete system.

- Steel: A 10-metre bay might be practical with appropriately sized steel beams.

Output

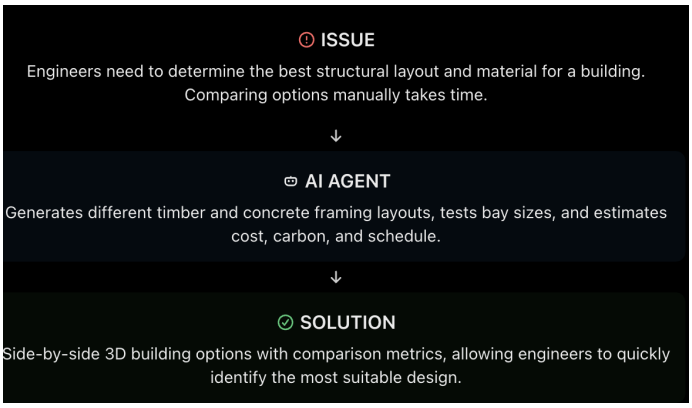
A comparison report (I guess our agents will be writing reports too)

And a 3D visual of each option, ideally a side-by-side walkthrough (for example, timber and concrete in split screen). Metrics: cost, embodied carbon, schedule and bay size.



This is a very simple conceptual visualization of what it could look like, our 3d models will be more complex and look similar to the first image in this doc.

AI slop image summary ->



Data used to create inputs and outputs:

Synthetic inputs and outputs until live access is available. Research outputs do not need to be correct;
WSP would plug in real data if the work were productized.

What Tom / WSP want

Tom (manager) suggested a chat interface beside a 3D workspace, similar to Claude's artifacts.

He also wants the team to assess whether a web-based or desktop approach makes more sense.

1. Agent architecture is our responsibility. Tom has deliberately left us to decide how the agents are organized and coordinated.
2. The main technical challenge is generating structural models. We need to figure out how agents interact with engineering tools and translate their outputs into usable 3D designs.
3. WSP is flexible about the deliverable. They value the research, system architecture, and working POC more than perfectly accurate results or an overly ambitious finished product.

ITEM	PHASE 1	PHASE 2
Accounts and network	No WSP email accounts and no WSP network or server access. Students use their own Queen's accounts and tools.	Unchanged: no WSP accounts or network access.
WSP APIs	No API source code. WSP shares API documentation and synthetic input and output data.	Live API access through a hosted endpoint in a new, externally accessible resource group: inputs in, results out, no code shared. IT lead time is several weeks.
Models and criteria	WSP shares an architectural model.	Real architectural model if available.
Autodesk Forma	Optional: a WSP Forma development project that students join with university emails, with WSP covering cloud and token costs.	Continues if used in Phase 1.
AI services and Revit	Students use their own AI tools. Revit student licences are free and not required if a cloud approach is used.	Possibly an AI service API key, with WSP covering token costs [budget TBD].

These details to be covered and clarified

Tools and tech stack:

Completely up to us what we want to use

We have to come up with:

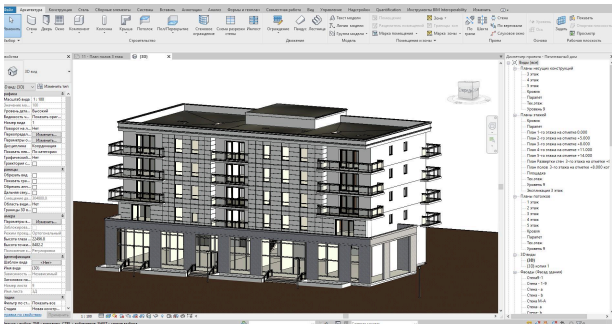
- Tech to use for AI agents
- Tech to use for front end/ backend
- Tech for 3d rendering

And API's/ data will be a mix of our own and whatever they provide to us. API's they provide -> idk yet but i assume its to calculate things like carbon emission, or calculate costs of a building based off the the model

Development: Use any LLM/ method of your choice

Possible AI orchestration: AI engineering team with orchestrator, each agent is very specialized in individual tasks, such as designing bay spacing etc.

AI agent flow:



- Take an architectural model (see example for what a model looks like).
- Generate and compare different models

(This part is repetitive i know but the image is cool)

Stretch goals (LATER WORRY)

Also a suggestion:

Designs could combine materials within the same building, such as a concrete podium with steel upper floors.

Once main pipeline/ product is successful and working, we have capacity to stretch and expand what we are building based off these option (LATER WORRY, dont need to understand all these yet)

OPTION	DESCRIPTION
Schedule vs material cost	Research the trade-off between construction schedule and material cost. Timber is an expensive material but fast to erect; concrete is cheap material but needs longer labour. No good industry comparison exists today. The output could be a 4D (time-based) model in the cloud that does not rely on Revit.
Location-based carbon	Carbon factors that include material procurement by location.
Cloud model automation	Use the Autodesk Design Automation API to build or modify the model in the cloud without opening Revit, and a Forma round-trip (the architect moves something and the design updates).
Platform-path recommendation	Guidance from the team on both paths: the Autodesk cloud (WSP's current ecosystem) versus a platform-agnostic web 3D viewer.
Conversational agent	A chat-based AI agent for interacting with the solution.
MEP and building science	AI auto-routing of ductwork to avoid clashes, and a schematic-stage operational energy model. Needs input from a WSP building-science subject matter expert.

Structural design comes first, with mechanical and energy options as possible extensions.